SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION with OpenCV   COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

1. Grey Level Transformation

(a) Purpose of Grey Level Transformation

Grey level transformation modifies the intensity values of an image to improve its appearance. It is used to:

- Enhance contrast
- Highlight important features
- Correct brightness
- Produce image negatives for better visualization

(b) Negative Transformation

Formula:

$$s = L - 1 - r$$

where:

- $L = 256$(for an 8-bit image)
- r= original pixel value

Example:

Original (r)	Negative ($s = 255 - r$)
50	205
100	155
150	105
200	55

2. Log Transformation

(a) Explanation

Log transformation expands dark pixel values and compresses bright pixel values.

Formula:

$$s = c\log(1 + r)$$

It improves details in dark regions while preventing bright areas from becoming too bright.

(b) Calculation

Given

$$\boldsymbol{c = 1}$$

For

$$r = 0,\ 1,\ 3,\ 7$$
$$s = \log(1 + r)$$

r	s
0	0
1	0.693
3	1.386
7	2.079

3. Histogram Processing

(a) Role of Histogram Equalization

Histogram equalization:

- Improves image contrast
- Spreads intensity values over the available range
- Makes hidden details more visible

(b) Equalization

Given 3-bit image

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels

$$N = 16$$

CDF

Intensity	Frequency	CDF
0	2	2
1	2	4
2	4	8
3	8	16

Equalized value

$$s = \frac{CDF}{N}(L-1)$$

where

$$L = 8$$

Intensity	CDF	Equalized
0	2	1
1	4	2
2	8	4
3	16	7

4. Spatial Filtering (Smoothing)

(a) Purpose

Smoothing:

- Reduces noise
- Removes small variations
- Blurs the image slightly
- Improves segmentation

(b) Averaging Filter

Example region

$$\begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$$

Average

$$= \frac{10 + 20 + 30 + 40 + 50 + 60 + 70 + 80 + 90}{9} = \frac{450}{9} = 50$$

New center pixel = 50

5. Spatial Filtering (Sharpening)

(a) Explanation

Sharpening:

- Enhances edges
- Improves fine details
- Increases contrast between neighboring pixels

(b) Laplacian

Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Example matrix

$$\begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$$

Output

$$\begin{aligned} &= 4(50) - 20 - 40 - 60 - 80 \\ &= 200 - 200 = 0 \end{aligned}$$

Center output = 0

6. Frequency Domain Filtering

(a) Low-pass vs High-pass

Low-pass	High-pass
Removes noise	Enhances edges
Blurs image	Sharpens image
Keeps low frequencies	Keeps high frequencies

(b)Given frequency components

$$[\mathbf{20},\ \mathbf{15},\ \mathbf{8}\ \mathbf{4}]$$

Low-pass retains first two

Filtered output

$$[\mathbf{20},\ \mathbf{15},\ \mathbf{0}\ \mathbf{0}]$$

7. Thresholding

(a) Global Thresholding

A single threshold value is applied to the whole image.

If

$$Pixel \geq T$$

→ 1

Otherwise

→ 0

(b)

Example

Pixels

$$[20,\ 45,\ 60\ 80]$$

Threshold

$$T = 50$$

Output

$$[0,\ 0,\ 1\ 1]$$

8. Edge Detection

(a) Explanation

Edge detection identifies sudden intensity changes in an image. It is useful for:

- Object detection
- Image segmentation
- Boundary extraction

(b) Sobel Horizontal

Mask

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Example image

$$\begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$$

Gradient

$$\begin{aligned} &= (-10 - 40 - 30) + (70 + 160 + 90) \\ &= -80 + 320 = 240 \end{aligned}$$

Gradient = 240

9. Region-Based Segmentation

(a) Region Growing

Region growing starts from a seed pixel and adds neighboring pixels whose intensity difference is within a specified threshold.

(b)

Example pixels

$$45,\ 48,\ 50,\ 52,\ 56,\ 60$$

Seed

$$50$$

Threshold difference

$$5$$

Accepted pixels

Difference from 50:

- 45 → 5 ✓
- 48 → 2 ✓
- 50 → 0 ✓

- 52 → 2 ✓
- 56 → 6 ✗
- 60 → 10 ✗

Region = {45, 48, 50, 52}

10. Combined Enhancement & Segmentation

(a) Why Smoothing Before Segmentation?

Smoothing:

- Removes noise
- Reduces false edges
- Produces more accurate segmentation
- Improves object boundaries

(b)

Given average value

$$48$$

Threshold

$$T = 50$$

Since

$$48 < 50$$

Segmented output

$$0$$

If multiple pixels are averaged, apply the same rule:

- Mean ≥ 50 → 1
- Mean < 50 → 0

Code of Conduct:

I V G ABBIRAMY – 192524097 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V G ABBIRAMY

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	**20%**	**Clearly interprets problem, identifies all parameters and requirements accurately**	**Understands problem with minor gaps**	**Partial understanding; misses key elements**	**Misinterprets or unable to understand problem**
Method Selection	**20%**	**Selects most appropriate method/formula with strong justification**	**Selects correct method with minor errors**	**Inappropriate or partially correct method**	**Unable to select method**

Solution Process	**25%**	**Logical, systematic steps with correct approach throughout**	**Mostly correct steps with minor mistakes**	**Disorganized steps; multiple errors**	**No clear process**
Accuracy of Calculation	**20%**	**All calculations accurate; correct final answer**	**Minor calculation errors**	**Frequent errors; incorrect final answer**	**No valid calculations**
Interpretation of Results	**15%**	**Clearly interprets results with units and engineering relevance**	**Basic interpretation with minor omissions**	**Limited or unclear interpretation**	**No interpretation**